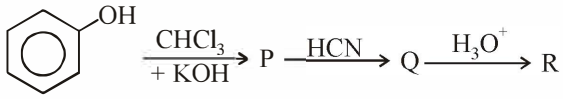
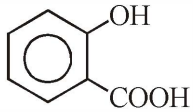
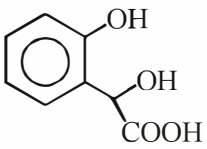
SOT TYPE

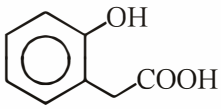
This section contains **12 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

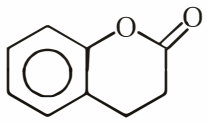
1. The major product(R) of the given reaction sequence is :



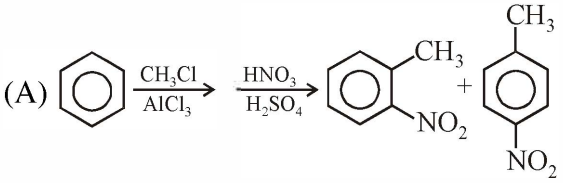
(A) 

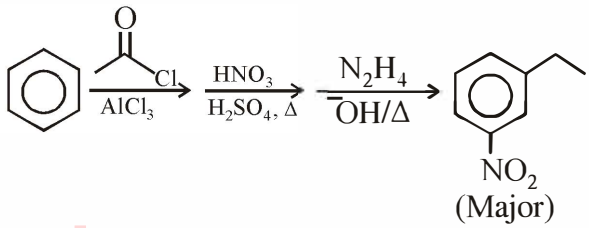
(B) 

(C) 

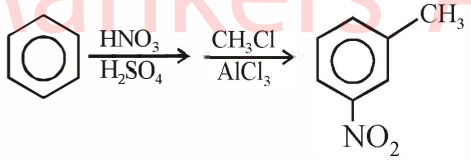
(D) 

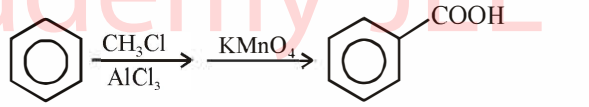
2. Which of given method is wrong to form the final product :

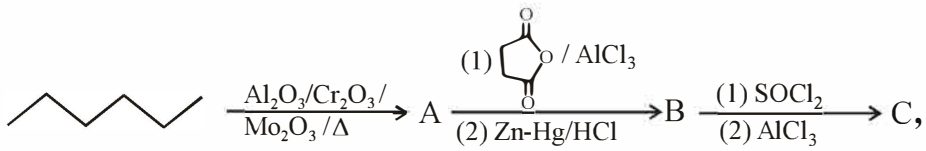
(A) 

(B) 

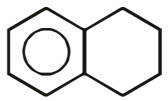
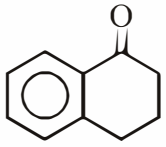
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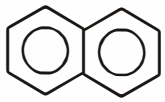
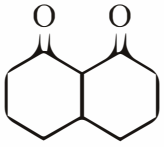
(C) 

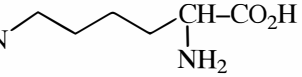
(D) 

3. 

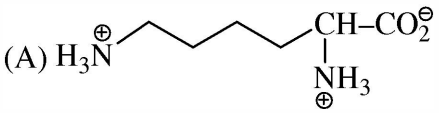
Identify 'C' product ?

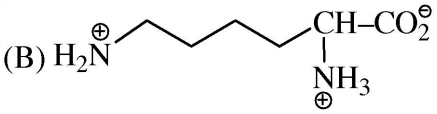
(A) 
(B) 

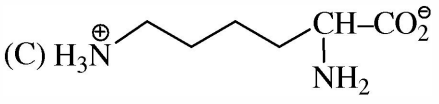
(C) 
(D) 

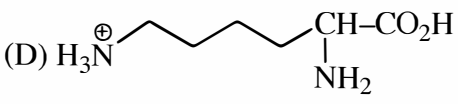
4. The correct dipolar ion structure for lysine having structure 

is :

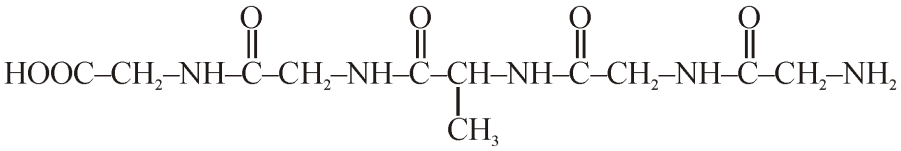
(A) 

(B) 

(C) 

(D) 

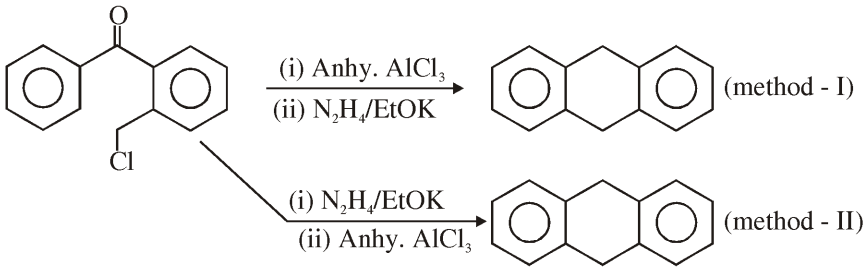
5. For a given compound



Incorrect statement is.

(A) It is tetrapeptide
(B) It is formed by condensation reaction
(C) On hydrolysis it give alanine & glycine
(D) It is optically active in nature.

6.



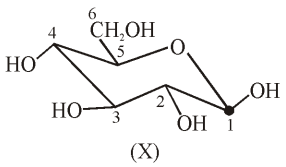
Correct statement :-

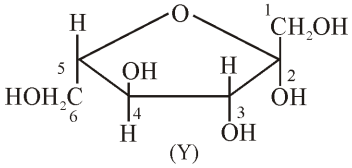
(A) Method I is more appropriate than II
(B) Method II is more appropriate than I
(C) Method I & II both are appropriate
(D) None is appropriate

7. Which one of the following statements is incorrect
(A) Vitamin B & C are water soluble
(B) Buna-S is addition co-polymer of 1,3-butadiene & styrene
(C) Uracil is purine base which is present in RNA
(D) All amino acids except glycine are optically active

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8. A disacahride (P) is obtained by condensation of X & Y, then choose correct option :


(X)


(Y)

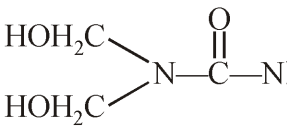
(A) If C_1 of (X) & C_2 of (Y) is used then (P) is reducing in nature
(B) (X) & (Y) both are reducing in nature & show mutarotation
(C) (X) is β-D (+) glucose & (Y) is α-D (–) fructose
(D) (X) is aldohexose & (Y) is α-pyranose

9. The addition polymer which is/are co-polymer :

(i) Neoprene (ii) Buna-N (iii) Nylon-66 (iv) SBR

(A) only iv (B) ii & iv (C) iii & iv (D) ii, iii & iv

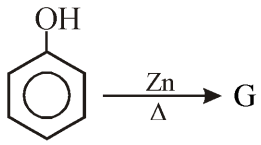
10. Polymer obtained by the given compound

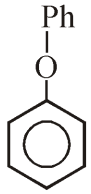


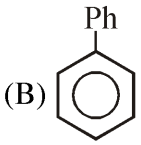
is :

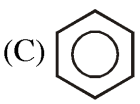
(A) Urea-formaldehyde resin
(B) Phenol formaldehyde resin
(C) Alkyd resin
(D) Melamine-formaldehyde resin

11. Identify G.

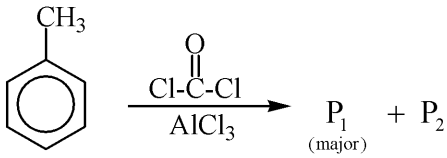


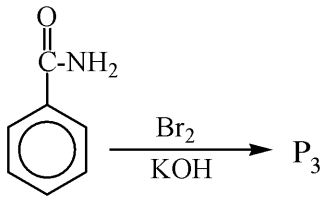
(A) 

(B) 

(C) 

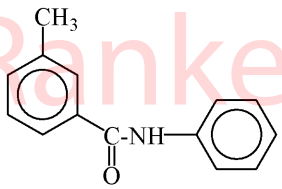
(D) 

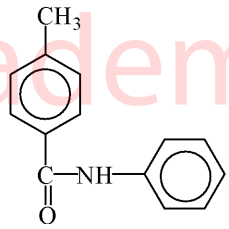
12. (i) 

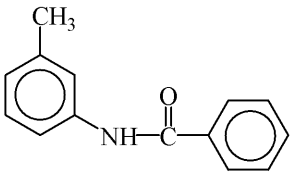
(ii) 

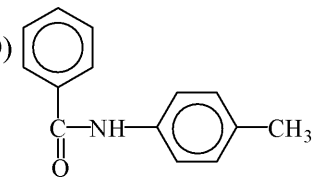
(iii) $P_1 + P_3 \longrightarrow P_4$

Identify P_4 (major product of (iii) reaction)

(A) 

(B) 

(C) 

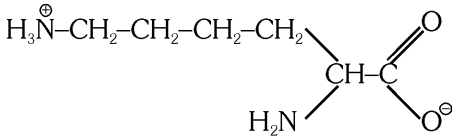
(D) 

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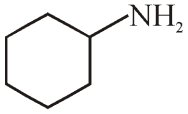
INTEGER TYPE

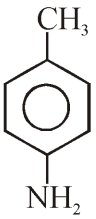
This section contains **3 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

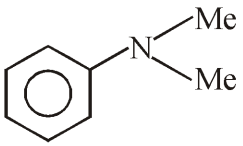
1. The total number of basic groups in the following form of lysine is :

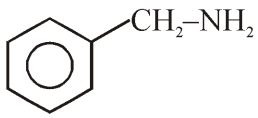


2. How many of the following compounds will show red colour dye formation with β-napthol after diazotisation

(i) 

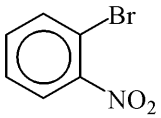
(ii) 

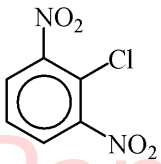
(iii) 

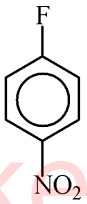
(iv) 

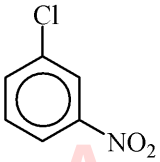
(v) 

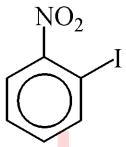
3. Number of compounds which can show faster rate of nucleophilic substitution of halogen than

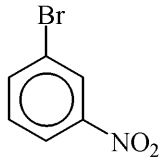


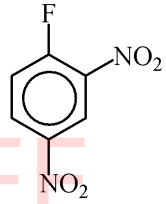
(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

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SOT TYPE

This section contains **13 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. The correct sequence of reagents used for conversion of Aniline into Benzylamine is ($PhCH_2–NH_2$)

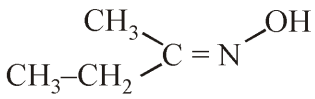
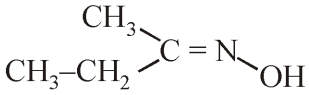
(A) $NaNO_2$/HCl, $Cu_2(CN)_2$/HCN, Sn/HCl (B) $NaNO_2$/HCl, CuCN, H_3O^+

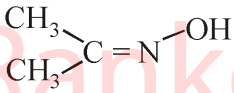
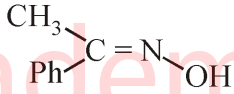
(C) $NaNO_2$/HCl, Cu^+/H_3PO_2, CH_3NH_2 (D) None

2. Ketoxime $\xrightarrow[\Delta]{PCl_5}$ A $\xrightarrow{H_3O^+}$ B + C $\xrightarrow[NaOH]{CHCl_3}$ CH_3NC

B $\xrightarrow{NaOH/CaO,\Delta}$ $CH_3–CH_3$

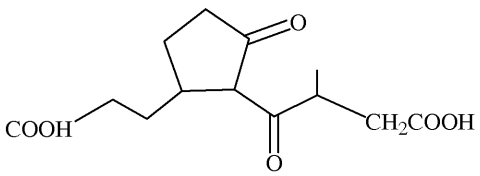
Identify ketoxime

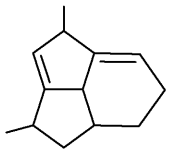
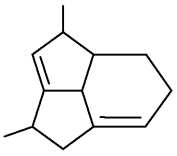
(A)  (B) 

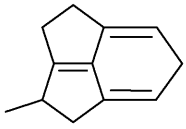
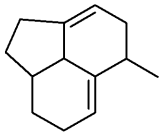
(C)  (D) 

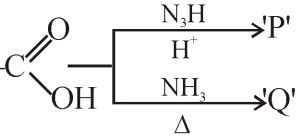
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3. Which molecule out of following will give the following dicarboxylic acid on heating with acidic solution of $KMnO_4$?



(A)  (B) 

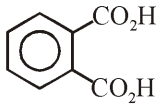
(C)  (D) 

4. P & Q are major product of following reaction 

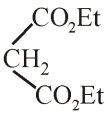
Reagent which convert 'Q' into 'P' is / are :

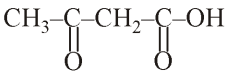
(A) Br_2 / NaOH (B) $NaNO_2$ / HCl (C) $H_3O^{\oplus}$ (D) None of these

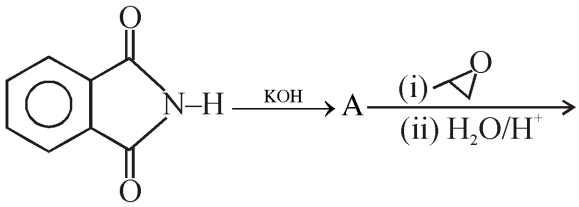
5. Which of the following compound loose CO_2 gas on heating ?

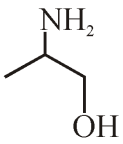
(A) 

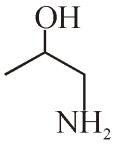
(B) CH_3CO_2H

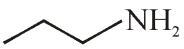
(C) 

(D) 

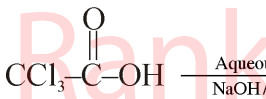
6.  major product is :

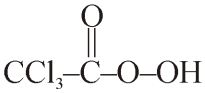
(A) 

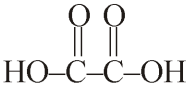
(B) 

(C) 

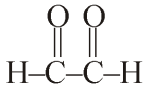
(D) 

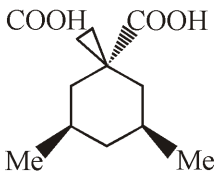
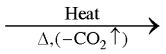
7.  product is ?

(A) 

(B) 

(C) $CHCl_3$

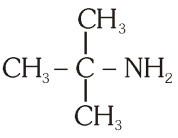
(D) 

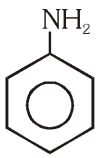
8.   A + B product ?

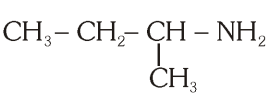
Identify relationship between A & B ?

(A) Enantiomers
(B) Diastereo isomers
(C) Functional isomers
(D) Positional isomers

9. Which of the following will not give carbylamine reaction

(A) 

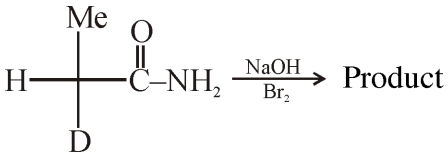
(B) 

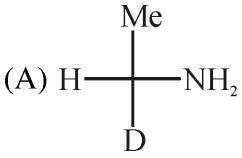
(C) 

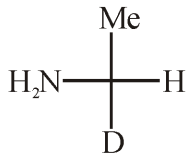
(D) $CH_3 - CONH_2$

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10 What will be correct product represented for the reaction,



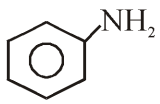
(A) 

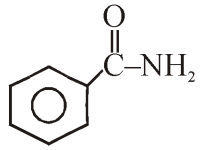
(B) 

(C) Mixture of both (A) and (B)

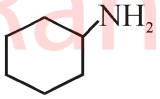
(D) Can not comment on stereochemistry of product

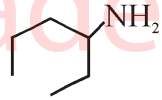
11. Which of the following compounds will not produce a compound with offensive smell when treated with $CHCl_3$/KOH ?

(A) 

(B) 

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(C) 

(D) 

12. **Statement-1 :** Gabriel phthalimide synthesis is used to prepare pure 1° aliphatic amines.

and

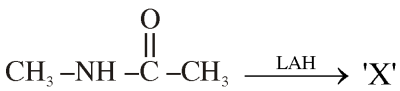
Statement-2 : 1° amines ($CH_3CH_2CH_2NH_2$) have higher boiling point than 3° amines $(CH_3)_3N$

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

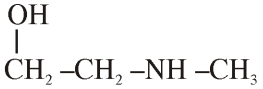
(D) Statement-1 is False, Statement-2 is True.

13. 

X is also obtained by reduction of which of the following using LAH

(A) 

(B) $CH_3 –CH_2 –NH –CH_3$

(C) 

(D) All of these

INTEGER TYPE

This section contains **3 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1. Compare the given characteristics of aniline and cyclohexanamine :

(i) Both are primary amine

(ii) Both can be acylated by RCOCl

(iii) Both reacts with $CHCl_3$/KOH

(iv) Both reacts with $NaNO_2$ + HCl at 0-5°C

(v) Both reacts with $PhSO_2Cl$ to give a compound which is soluble in KOH

(vi) Both gives coupling reaction with phenol

(vii) Both gives electrophilic substitution reaction

How many of the given characteristics are correct?

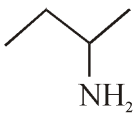
2. Number of reagents which can convert aliphatic aldehyde into carboxylic acid or its salt.

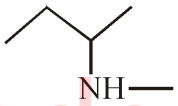
(1) conc. OH^- (2) CF_3–COOH (3) $[Ag(NH_3)_2]^+ OH^-$ (4) $HgCl_2$

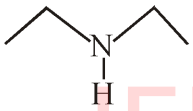
(5) CH_3–COOH (6) $K_2Cr_2O_7$, H^+, Δ (7) $KMnO_4$, H^+, Δ (8) Anhydrous H_2CrO_4

3. How many of the following compounds will give isocyanide (carbylamine) test positive :

(a) 

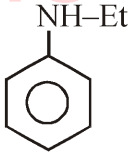
(b) 

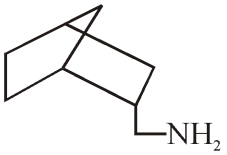
(c) 

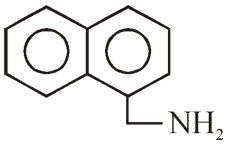
(d) 

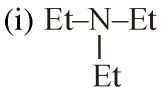
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(e) 

(f) 

(g) 

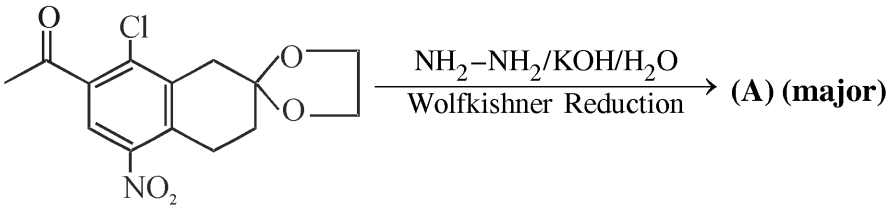
(h) 

(i) 

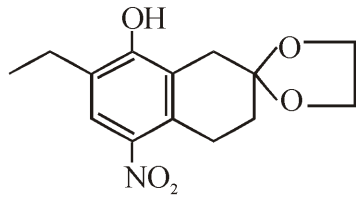
SOT TYPE

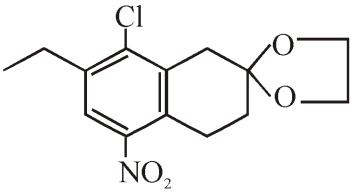
This section contains **13 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

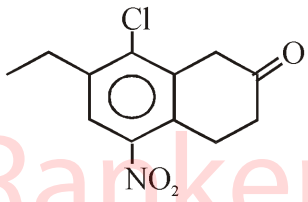
1. Predict the major product of the **given reaction :-**



(A) (major)

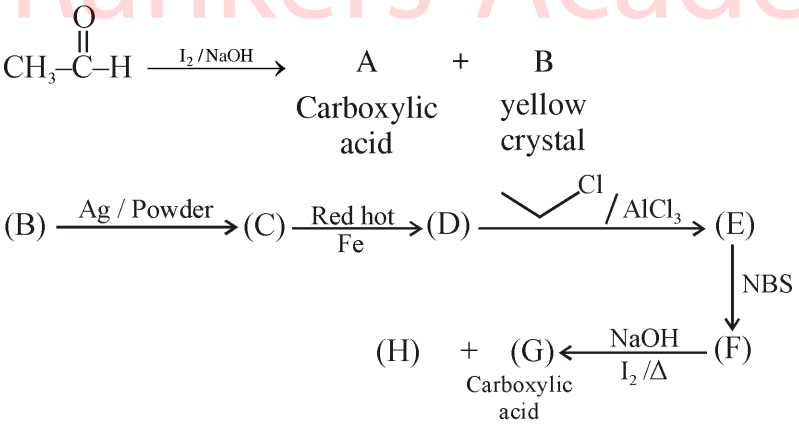
(A) 

(B) 

(C) 

(D) 

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2. 

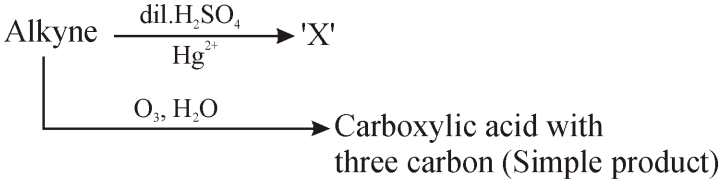
Correct staement for above sequence

(A) (A) & (G) are same

(B) (B) & (H) are same

(C) (F) is a Ketone

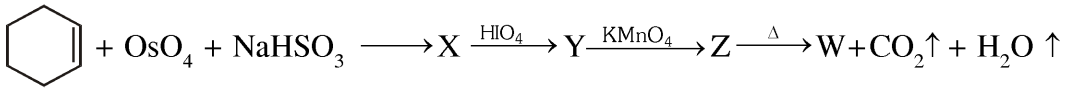
(D) Reaction sequence involve haloform reaction & friedel craft Acylation

3. 

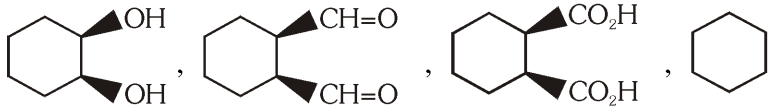
Correct statement regarding 'X' is -

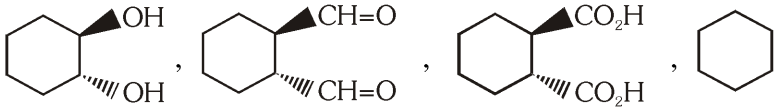
(A) Aldehyde (B) Acid (C) Alcohol (D) 3-hexanone

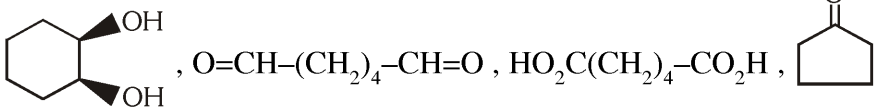
1001CJA00CHEN2500301

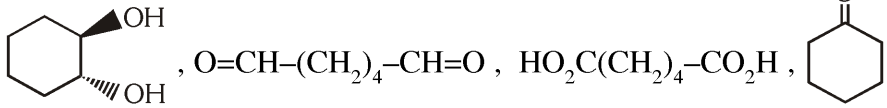
4. 

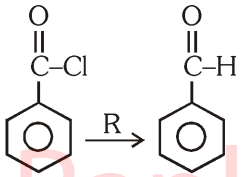
X, Y, Z, W are respectively -

(A) 

(B) 

(C) 

(D) 

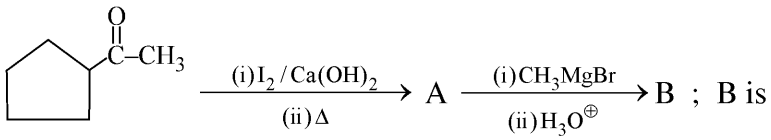
5. 

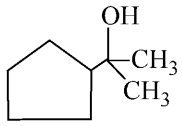
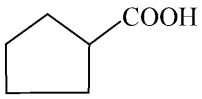
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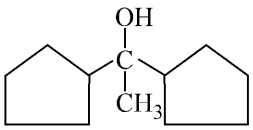
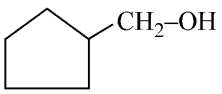
Which of the following can be used as R in the above reaction :

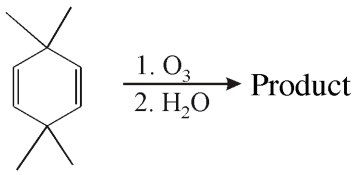
(A) $LiAlH_4$ (B) Ni / H_2

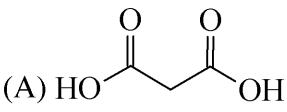
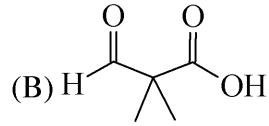
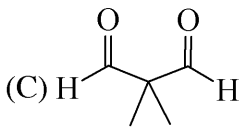
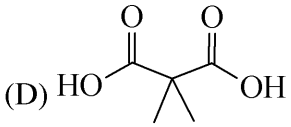
(C) $Pd / BaSO_4 / CaCO_3 / H_2$ (D) $NaBH_4$

6.  ; B is

(A)  (B) 

(C)  (D) 

7. 

(A)  (B)  (C)  (D) 

8. Which one of the following reaction will not produce hydrocarbon as final product -

(A) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[(ii)\ KOH,\Delta]{(i)\ NH_2-NH_2}$

(B) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow[300^\circ C]{Zn\text{-}Hg(HCl)}$

(C) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H \xrightarrow[\Delta]{NaOH}$

(D) $CH_3-CH=O \xrightarrow[(ii)\ Red\ P/HI]{(i)\ H_2/Ni}$

9. Most reactive towards Nucleophilic attack -

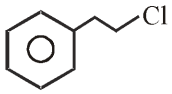
(A) CH_3CHO

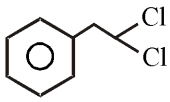
(B) CH_3COCl

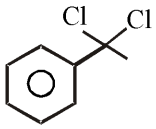
(C) $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3$

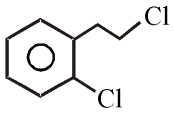
(D) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

10. Which of the following molecule on alkaline hydrolysis produces a compound which can show Iodoform test ?

(A) 

(B) 

(C) 

(D) 

11. The correct statement for the given reaction is

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$$\begin{array}{l} CH_3 \\ Br-H \\ Br-H \\ CH_3 \end{array} \xrightarrow[Acetone]{NaI} X \xrightarrow{CF_3CO_3H} Y$$

(A) Y is optically inactive due to external compensation

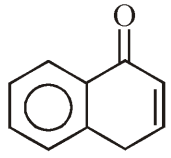
(B) Y is optically inactive due to internal compensation

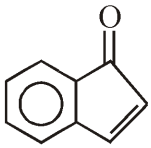
(C) X is predominantly cis-alkene

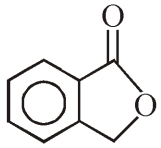
(D) Y does not have chiral centres

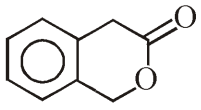
12. The final product is :

$$\text{o-phthalaldehyde} \xrightarrow[\Delta]{OH^-} P \xrightarrow[\Delta]{H^+} Q\ (Final)$$

(A) 

(B) 

(C) 

(D) 

13. $CH_3CH_2CH_2-\underset{\underset{CH_3}{|}}{CH}-CH=O \xrightarrow{HCHO,\ OH} A(C_7H_{14}O_2)$

$A(C_7H_{14}O_2) \xrightarrow{HCHO\ /O\bar{H}} B\ (C_7H_{16}O_2)$ (formed as one of the product)

'B' is

(A) $CH_3-CH_2-CH_2-\underset{\underset{CH_2OH}{|}}{CH}-CH_2CH_2OH$

(B) $CH_3-CH_2-CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_2-OH}{|}}{C}}-CH_2OH$

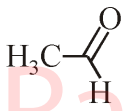
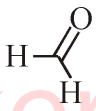
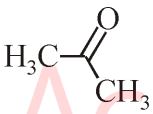
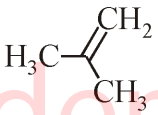
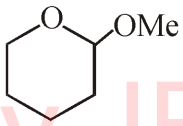
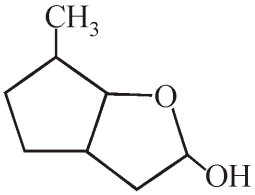
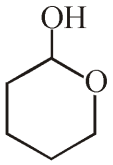
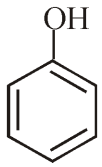
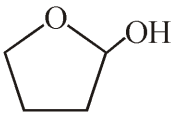
(C) Both

(D) None

INTEGER TYPE

This section contains **2 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1. Identify total number of compounds give positive test with tollens reagent (silver mirror test) :

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2. $R-\underset{\underset{O}{\|}}{C}-R \rightarrow RCH_2R$

Identify number of reagents that can be used for above conversion :-

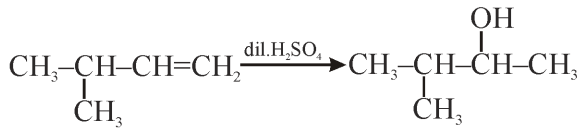
(a) $Zn(Hg)$ + conc. HCl (b) $CHCl_3$ + NaOH (c) $LiAlH_4$ (d) $NaBH_4$

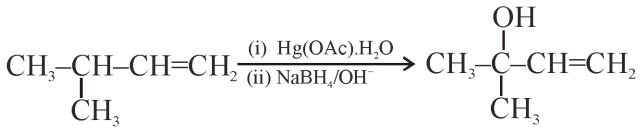
(e) $N_2H_4 + \bar{O}H$ (f) SeO_2 (g) MnO_2 (h) $\underset{\underset{CH_2SH}{|}}{\overset{\overset{CH_2SH}{|}}{}}$, H_2 (Raney Ni)

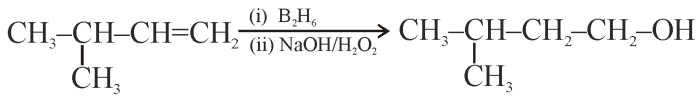
SOT TYPE

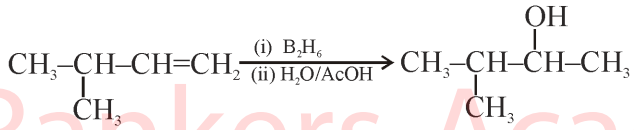
This section contains **15 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. Which of the following represent correct major product formation -

(A) 

(B) 

(C) 

(D) 

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2. Consider the following reaction :

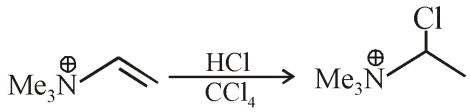
$C_2H_5OH + H_2SO_4 \rightarrow$ Product

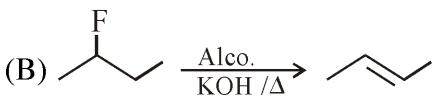
Among the following, which one cannot be formed as a product under any conditions ?

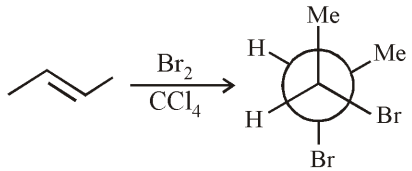
(A) Ethyl-hydrogen sulphate (B) Ethylene

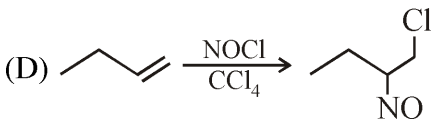
(C) Acetylene (D) Diethyl ether

3. Identify reactions correctly match with their major product .

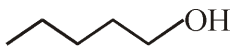
(A) 

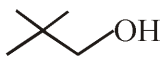
(B) 

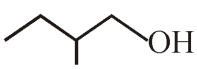
(C) 

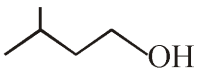
(D) 

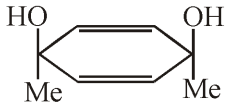
4. Which of the following alcohols cannot be prepared from an alkene ?

(A) 

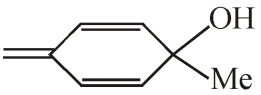
(B) 

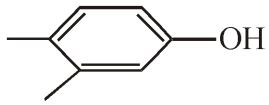
(C) 

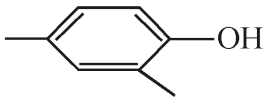
(D) 

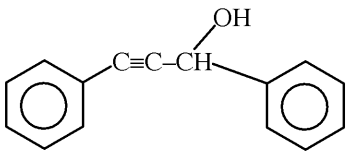
5.  $\xrightarrow[\Delta]{H^+}$, Major product -

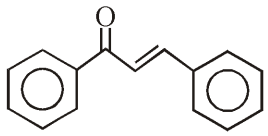
(A) 

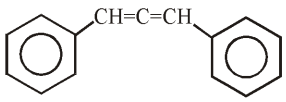
(B) 

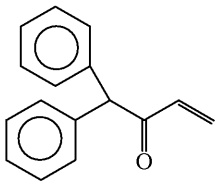
(C) 

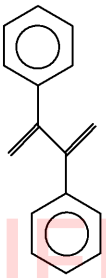
(D) 

6.  $\xrightarrow{H^+}$ Product is :

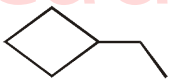
(A) 

(B) 

(C) 

(D) 

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7. Number of monochlorinated products when  undergoes reaction with $Cl_2/h\nu$

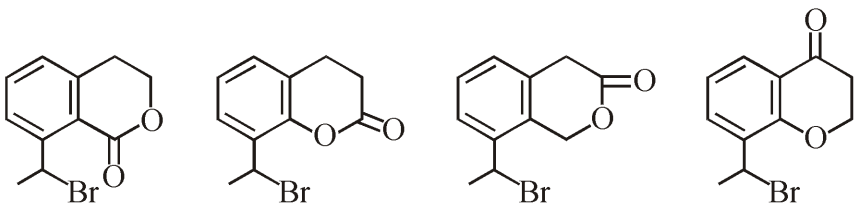
(A) 6 (B) 8 (C) 10 (D) 12

8. $CH_3CH_2—L \longrightarrow CH_3CH_2—Cl$

The above conversion is perfomed in a weakly polar aprotic solvent, such as acetone. Which of the following substance is best to proceed reaction fastest ?

(A) Li^+Cl^- (B) Na^+Cl^- (C) $Bu_4N^+Cl^-$ (D) HCl

9. Consider the following molecules :

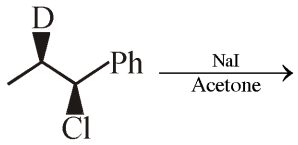


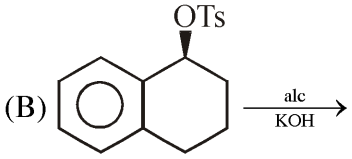
I II III IV

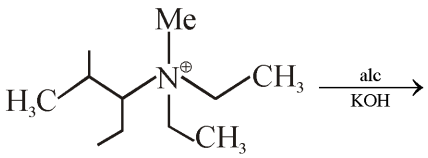
The correct decreasing ease of reaction with moist Ag_2O :

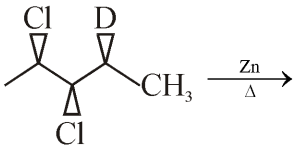
(A) II > III > IV > I (B) II > IV > III > I (C) II > I > III > IV (D) IV > II > III > I

10. Which of the following reaction will not give the chiral product:

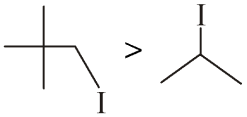
(A) 

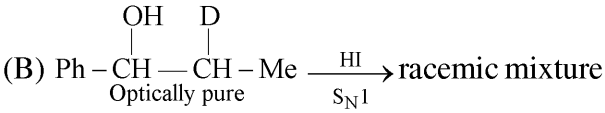
(B) 

(C) 

(D) 

11. Choose the correct option.

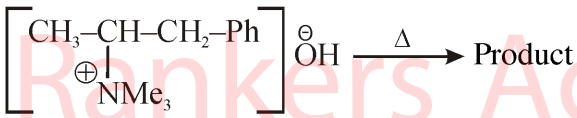
(A) Rate of S_N2 

(B) 

(C) $Me-Cl \xrightarrow[Ag_2O]{Moist} Me-O-Me$ (major)

(D) $MeI\ (excess) + NH_3 \longrightarrow Me_4\overset{\oplus}{N}$

12. Which of the given statement is incorrect about the following reaction -



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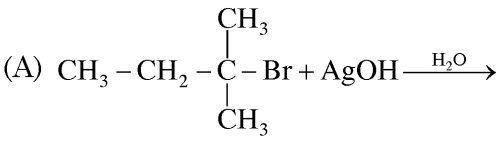
(A) Major product is hoffmann's alkene

(B) Major product is decided by more stable β-anion

(C) Mechanism of major product formation is E_{1CB}.

(D) Reaction is known as 'Pyrolysis of quaternary ammonium hydroxide'.

13. Which of the following reaction is expected to take place by S_N2 mechanism ?

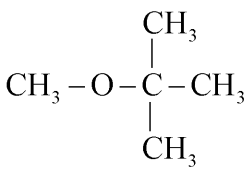
(A) 

(B) $(CH_3)_3C-Br + KOH \xrightarrow{H_2O}$

(C) $CH_3CH_2CHBrCH_3 + KOH \xrightarrow[heat]{C_2H_5OH}$

(D) $(CH_3)_2CHBr + CH_3COONa \xrightarrow{DMF}$

14. The ether that cannot be synthesized by Williamson ether synthesis method.

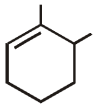
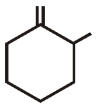
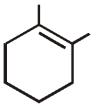
(A) 

(B) $PhOCH_2CH_3$

(C) $CH_3 - CH_2 - O - CH_2CH_3$

(D) $(CH_3)_3C-O-C(CH_3)_3$

15. The relative yield of the following alkenes produced by the reaction of 1-chloro-(Z)-1, 2-dimethylcyclohexane with KOH/heat is

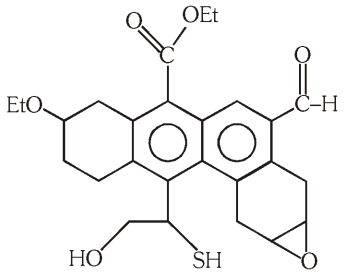
(i)  (ii)  (iii) 

(A) i > ii > iii (B) ii > iii > i (C) i > iii > ii (D) iii > i > ii

INTEGER TYPE

This section contains **2 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1. How many moles of Grignard reagent are consumed in given compound.



2. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_3$ (X)

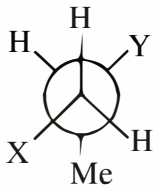
when X undergoes monochlorination and then the products fromed undergo fractional distillation, the number of fractions obtained is :

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SOT TYPE

This section contains **12 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

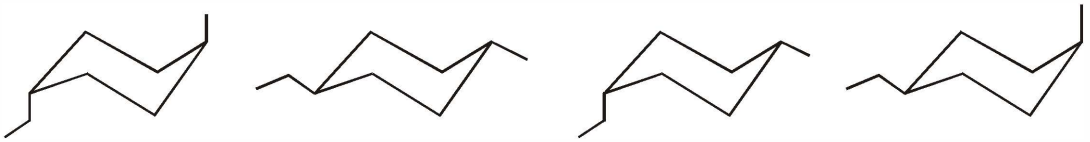
1. Newmann projection representation of $CH_3–CH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–Et$ about $C_2 - C_3$ is



Then X & Y are respectively :

(A) Me, Et (B) H, Et (C) Et, H (D) Et, Me

2. Which of the following represent correct stability order of

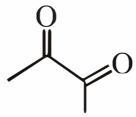
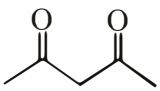
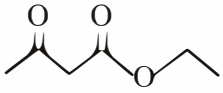
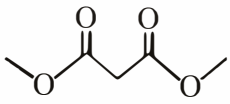


(I) (II) (III) (IV)

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(A) II > I > IV > III (B) II > III > IV > I (C) II > IV > III > I (D) II > IV > I > III

3. Which compound predominantly exist in its enol form -

(A)  (B)  (C)  (D) 

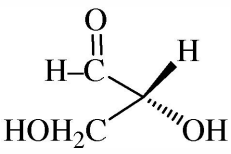
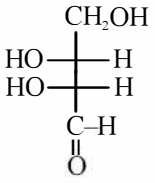
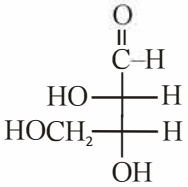
4. Total number of possible cyclic isomer of molecular formula C_5H_{10} is

(A) 7 (B) 6 (C) 5 (D) 4

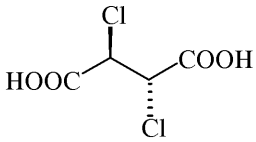
5. 4 gm of (+)-2-chlorobutane is mixed with 6 gm of (–)2 chlorobutane . What will be the optical purity of the obtained mixture.

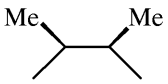
(A) 10% (B) 20% (C) 30% (D) 40%

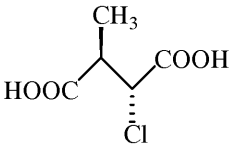
6. Identify compounds having 'D' configuration ?

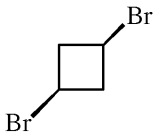
(A)  (B)  (C)  (D) All of these

7. Identify meso compound -

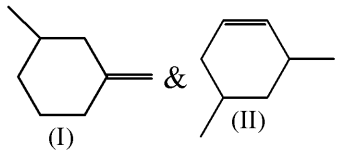
(A) 

(B) 

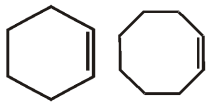
(C) 

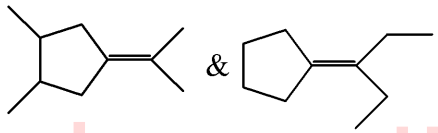
(D) 

8. Select the pair in which first member show geometrical isomerism but not second member-

(A) 

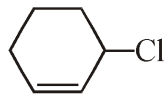
(B) 

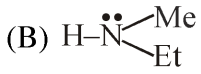
(C) 

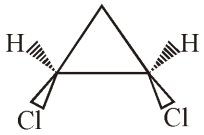
(D) 

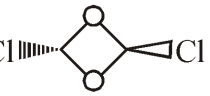
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9. The resolvable compound among following

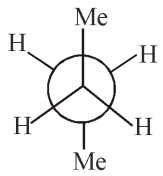
(A) 

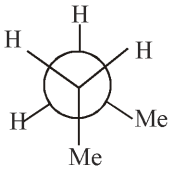
(B) 

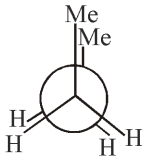
(C) 

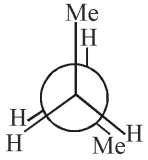
(D) 

10. Which of the following is most stable conformation of n-butane about (C_2-C_3) bond-

(A) 

(B) 

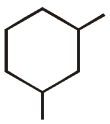
(C) 

(D) 

11. Identify total number of possible ether for molecular formula $C_4H_{10}O$ [Excluding Stereo Isomers].

(A) 4 (B) 3 (C) 5 (D) 6

12. How many stereoisomers are possible for the following molecule ?

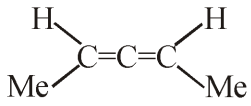


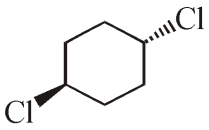
(A) 1 (B) 2 (C) 3 (D) 4

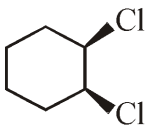
INTEGER TYPE

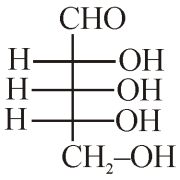
This section contains **3 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

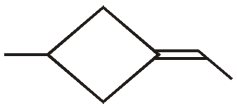
1. How many compounds from the following are optically active.

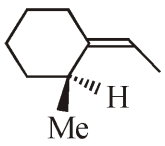
(a) 

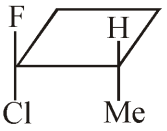
(b) 

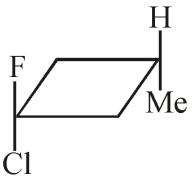
(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

2. How many structural isomers & stereo isomers are possible with given following molecular formula

(i) $C_5H_{10}O$ — (Total Aldehyde) / (Total Ketone)

(ii) C_4H_8

(iii) $C_4H_{10}O$ (Total Alcohol)

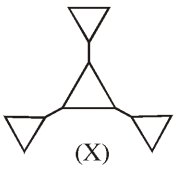
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(iv) $C_4H_{11}N$

(v) C_5H_{10} — Open chain / Close chain

(vi) $C_5H_8O_2$ (Total carboxylic acid)

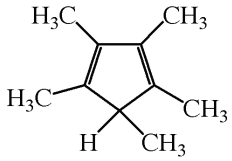
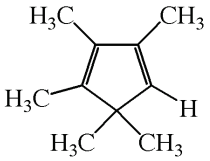
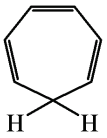
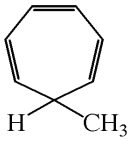
3. Number of geometrical isomers of X = b.
Number of structural isomers formed when same compound X undergo monochlorination = a
Find out a + b.


(X)

SOT TYPE

This section contains **12 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. Which one of the hydrocarbon is the most acidic

(A)  (B)  (C)  (D) 

2. The correct order of stability.

(A) $\overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-NH_2 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-OCH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-H > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_2OCH_3$

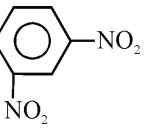
(B) $\overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_2-OCH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-H > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-NH_2 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-OCH_3$

(C) $\overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-H > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_2-OCH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-NH_2 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-OCH_3$

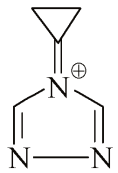
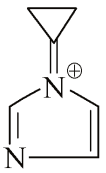
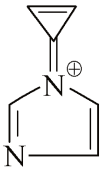
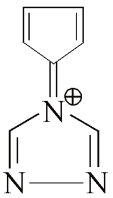
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(D) $\overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_2-OCH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-H > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-CH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-OCH_3 > \overset{\ominus}{C}H_2-\underset{\underset{O}{\|}}{C}-NH_2$

3. Which carbanion is stable due to pπ-pπ bonding :-

(A)  (B) $Ph_3\overset{\oplus}{P}-\overset{\ominus}{C}H_2$ (C) $RS-\overset{\ominus}{C}H-SR$ (D) $CH_3C\equiv\overset{\ominus}{C}$

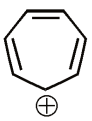
4.

 (a)
 (b)
 (c)
 (d)

Correct order of basic strength of this compound is

(A) d > b > c > a (B) c > b > a > d (C) c > a > b > d (D) a > d > c > b

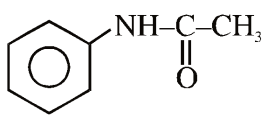
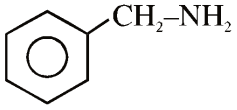
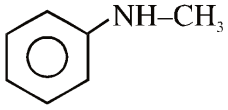
5. Identify aromatic species ?

(A)  (B)  (C)  (D) (A) & (B) are correct

6. Identify Lewis acidic intermediate ?

(A) $\dot{C}H_3$ (B) $\overset{\oplus}{C}H_3$ (C) $\overset{\ominus}{C}H_3$ (D) All of these

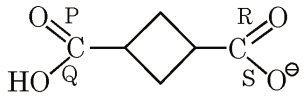
7. For given compounds.

(P) (Q) (R)

Basic strength order is.

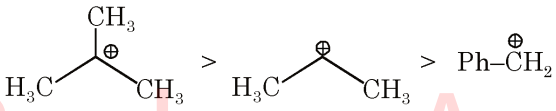
(A) P > Q > R (B) Q > R > P (C) R > Q > P (D) R > P > Q

8. 

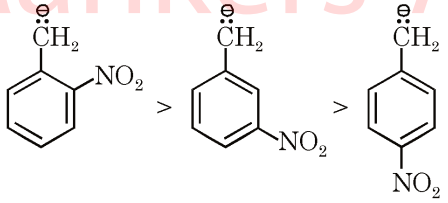
Select correct order of bond length of above bond P, Q, R & S :

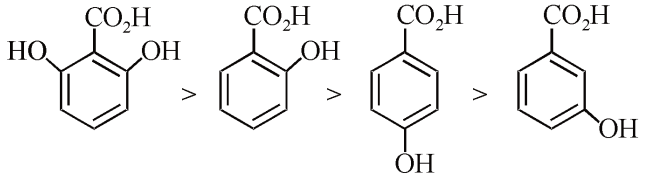
(A) P > R > S > Q (B) Q > R = S > P (C) S > Q > R > P (D) None of these

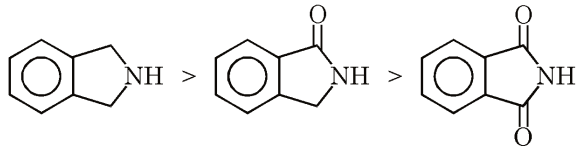
9. Find out the correct statement :

(A)  (Order of stability of carbocation)

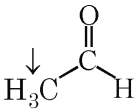
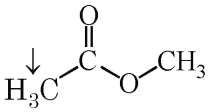
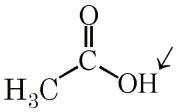
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(B)  (Order of stability of carbanion)

(C)  (Order of acidic strength)

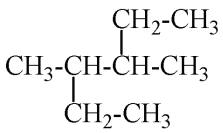
(D)  (Order of basic strength)

10. The relative acidity of the indicated H in each of the following is

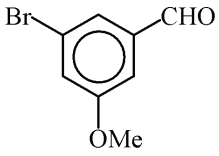
(i)  (ii)  (iii) 

(A) i > ii > iii (B) ii > iii > i (C) i > iii > ii (D) iii > i > ii

11. IUPAC name of compound



(A) 2, 3-diethyl butane
(B) 2-ethyl-3-methyl pentane
(C) 3-methyl-2-ethyl pentane
(D) 3,4-dimethyl hexane

12. Correct IUPAC name for the given compound 

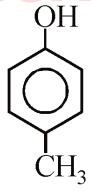
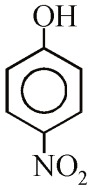
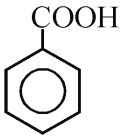
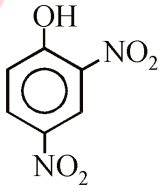
(A) 3-Methoxy-5-bromo benzene carbaldehyde
(B) 3-Formyl-5-bromo phenyl methyl ether
(C) 3-Formyl-5-bromo-1- methoxy benzene
(D) 3-Bromo-5-methoxy benzaldehyde

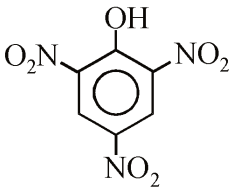
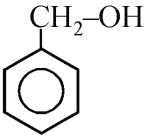
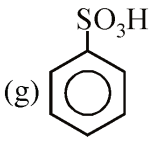
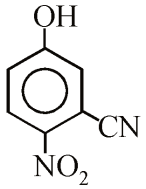
INTEGER TYPE

This section contains **2 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

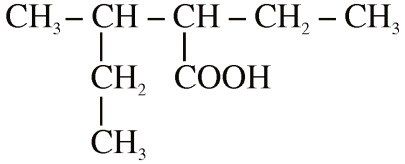
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1. How many of the following phenol derivatives produces efference CO_2 with $NaHCO_3$.

(a) 
(b) 
(c) 
(d) 

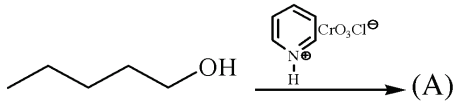
(e) 
(f) 
(g) 
(h) 

2. Number of carbon atoms in the principal carbon chain of the given compound in its IUPAC name will be.

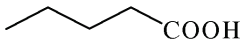
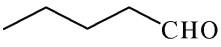
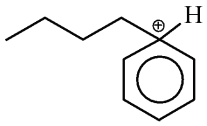
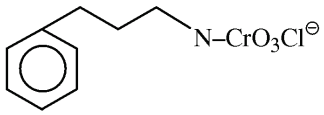


SOT TYPE

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. 

(A) is defined as

(A) 
(B) 
(C) 
(D) 

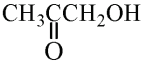
2. What will be the major product of following reaction.



(A) 
(B) 
(C) 
(D) 

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3. Fehling's solution can make distinction between:

(A) CH_3CHO and C_6H_5CHO
(B) CH_3CHO and 
(C)  and HCHO
(D) CH_3CHO and HCHO

4. Acetone + $PCl_5 \longrightarrow P_1 + POCl_3$

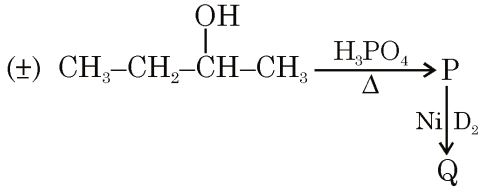
$P_1 + NaNH_2$ (excess) $\longrightarrow P_2$

$P_2 + CH_3–I \longrightarrow P_3$

P_3 is :

(A) But-1-yne (B) Pent-1-yne (C) But-2-yne (D) Pent-2-yne

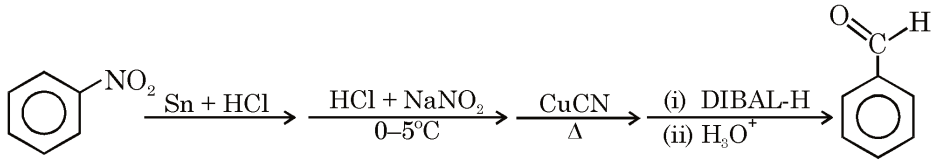
5. If (P) & (Q) are major products, then find incorrect statement for given sequence :

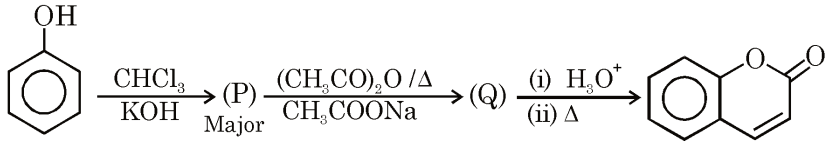


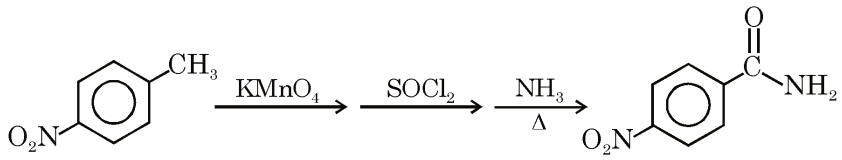
(A) Formation of (Q) is syn addition phenomenon
(B) Formation of (P) is unimolecular elimination reaction
(C) Formation of (Q) is redox reaction
(D) (Q) is optically inactive due to internal compensation

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6. Incorrect reaction sequence is :

(A) 

(B) 

(C) 

(D) $CH_3–CH=CH–COOH \xrightarrow[Ni]{H_2} (P) \xrightarrow[CaO/\Delta]{NaOH} CH_3CH_3$

7. Select the correct statement among following :

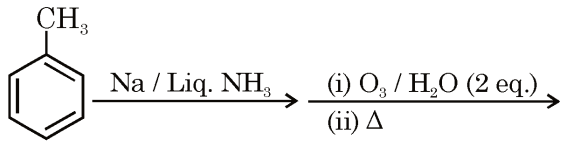
(A) Dimethylamine on reaction with Hinsberg reagent form compound which is insoluble in aq. NaOH

(B) Proline is the amino acid which is chiral and have 1° amine group

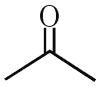
(C) Dimethylamine can be prepared by Gabriel phthalmide synthesis method

(D) Sucrose is sugar which reduces Tollen's reagent and also show mutarotation

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8. 

Final product(s) obtained in above reaction is/are :

(A) 

(B) $CH_3–COOH$ (C) CO_2 (D) All of these

9. Identify the correct sequence for True and False

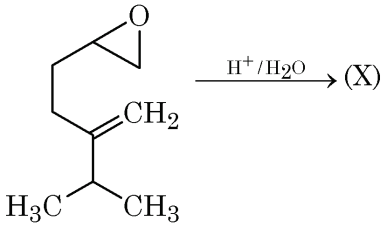
(I) Only compounds with chiral centers can be optically active

(II) Presence of elements of symmetry is the reason for a molecule's optical activity

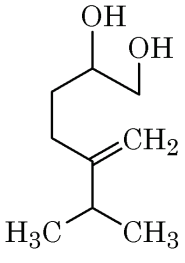
(III) All organic compounds contain only carbon & hydrogen

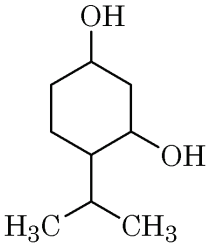
(IV) Resonating structures are real structures.

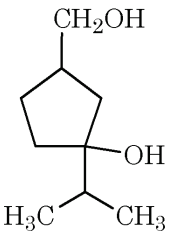
(A) TTFT (B) FFFF (C) TTTT (D) TTFF

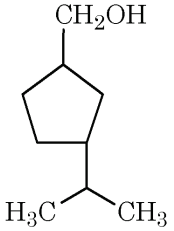
10. 

The product (X) is :

(A) 

(B) 

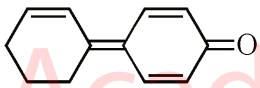
(C) 

(D) 

INTEGER TYPE

This section contains **5 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1. Total number of hydrogen replaced by deuterium in enol form of compound (M), when it is treated with NaOD / D_2O for long time ?



(M)

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2. Find the total number of substrates more reactive than $CH_3CH_2CH_2CH_2Cl$ with KI in acetone.

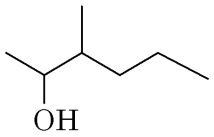
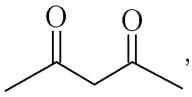
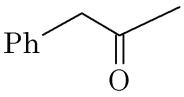
$CH_3Cl, (CH_3)_2CHCl, CH_2=CH-CH_2Cl, PhCH_2Cl, CH_3OCH_2Cl, PhCOCH_2Cl, (CH_3)_3CCl, C_6H_5Cl$

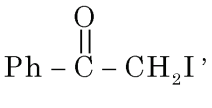
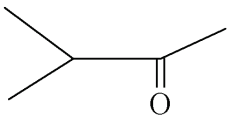
3. How many of the given are condensation co-polymers :

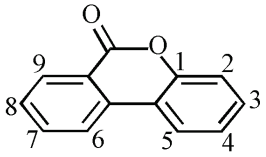
(i) BUNA-S (ii) BUNA-N (iii) Nylon-6 (iv) Dacron (v) Neoprene

(vi) PHBV (vii) Nylon-2-Nylon-6 (viii) Teflon (ix) Bakelite (x) Cellulose

4. Identify the total number of compounds that gives positive iodoform test

 ,  ,  , Lactic acid, Ethanol, Chloral,

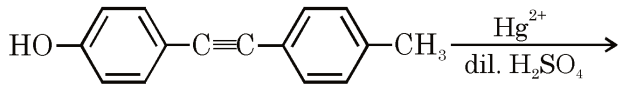
 ,  , CH_3-NO_2

5.  products.

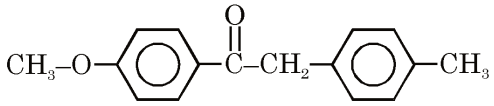
major organic product will have Bromine at position.

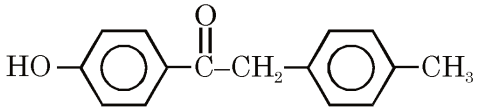
SOT TYPE

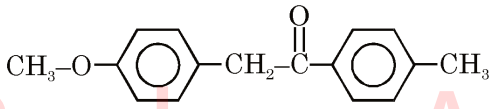
This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. 

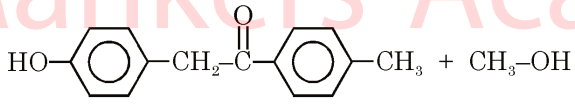
Major product is :

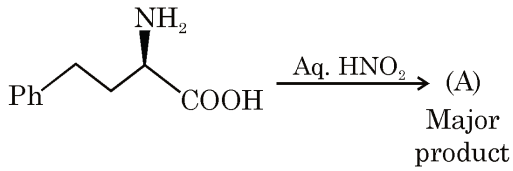
(A) 

(B) 

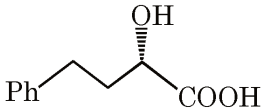
(C) 

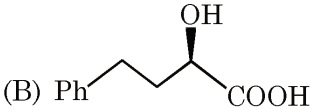
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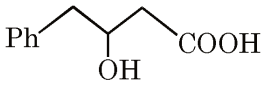
(D) 

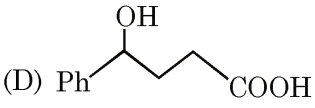
2. 

Major product of above reaction is :

(A) 

(B) 

(C) 

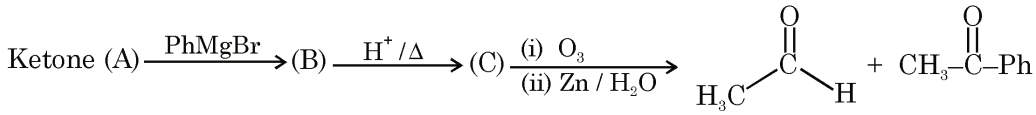
(D) 

3. Which of the following is natural condensation Co–polymer ?

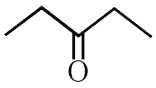
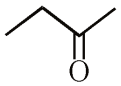
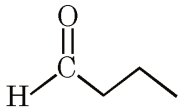
(A) Natural rubber (B) Starch

(C) Insulin (D) Dacron

4. Consider the following reaction sequence :



The ketone (A) is :

(A)  (B)  (C)  (D) 

5. Incorrect statement :

(A) Vinyl chloride > Benzyl chloride (C–Cl bond length)

(B) Allyl bromide > Vinyl bromide (Reactivity towards S_N2)

(C) Tertiary butyl bromide > Secondary butyl bromide (Reactivity towards S_N1)

(D) 7-Chloro 1,3,5-cycloheptatriene > 5-chloro 1,3-cyclopentadiene (Rate of hydrolysis)

6. Which of the following statements is correct ?

(A) Lysine is example of basic & nonessential amino acid

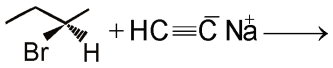
(B) The IUPAC name of isoprene is 2-chloro-1,3-butadiene

(C) Lactose is example of disacchride which is formed by combination of C_1 of β-D-galactose & C_4 of β-D-glucose followed by elimination of H_2O

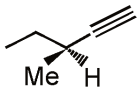
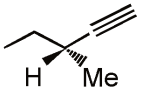
(D) Glucose gives +ve test with Tollen's ; Fehling ; schiff reagent & $NaHSO_3$ solution

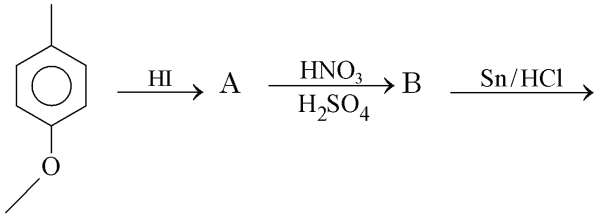
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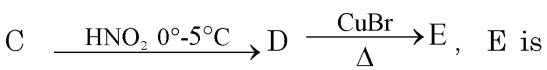
7. Consider the following reaction.



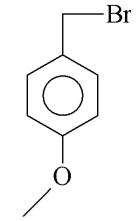
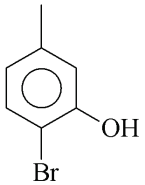
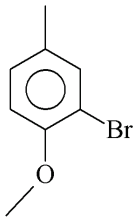
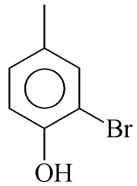
Which of the following products is not expected to be formed ?

(A)  (B)  (C)  (D) 

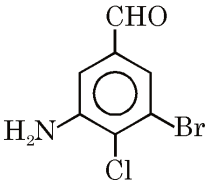
8. 



, E is

(A)  (B)  (C)  (D) 

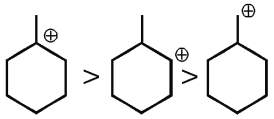
9. Correct IUPAC name of following compound is :

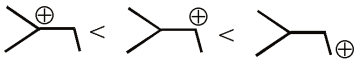


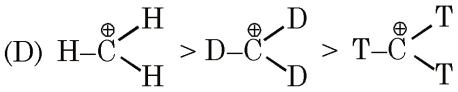
(A) 5-amino-3-bromo-4-chloro benzaldehyde
(B) 3-amino-4-bromo-5-chloro benzaldehyde
(C) 5-amino-4-bromo-3-chloro benzaldehyde
(D) 3-amino-5-bromo-4-chloro benzaldehyde

10. Identify correct stability order of carbocation ?

(A) $CH_3^{\oplus} > CH_3-CH_2^{\oplus} > CH_3-\overset{\oplus}{C}H-CH_3$

(B) 

(C) 

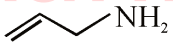
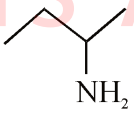
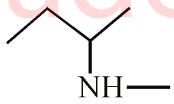
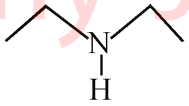
(D) 

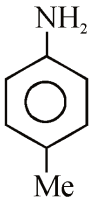
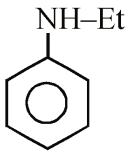
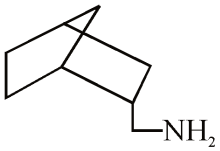
INTEGER TYPE

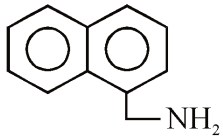
This section contains **3 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

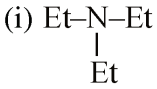
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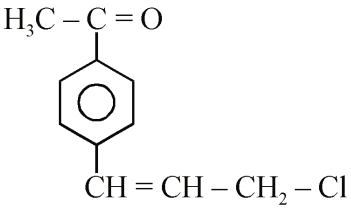
1. How many of the following compounds will give isocyanide (carbylamine) test positive :

(a) 
(b) 
(c) 
(d) 

(e) 
(f) 
(g) 

(h) 

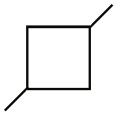
(i) 

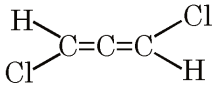
2. 

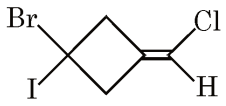
How many reactions are possible for the above compound?

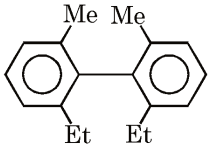
(i) Nucleophilic substitution
(ii) Free radical substituion
(iii) Electrophilic addition
(iv) Nucleophilic addition
(v) Aldol condensation
(vi) Electrophilic substitution

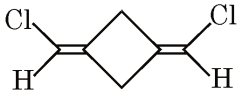
3. How many compounds show geometrical isomerism ?

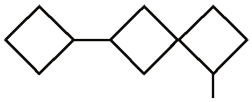
(a) 

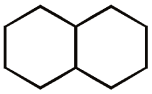
(b) 

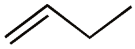
(c) 

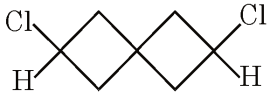
(d) 

(e) 

(f) 

(g) 

(h) 

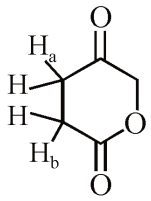
(i) 

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SOT TYPE

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. The correct acidic order of H

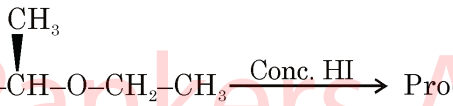


(A) $H_a > H_b$ (B) $H_b > H_a$ (C) $H_a = H_b$ (D) Can not decided

2. $CH_3CHO \xrightarrow[H^+]{MnO_4^-} A \xrightarrow{SOCl_2} B \xrightarrow[\Delta]{CH_3CO_2Na} C$. In the above reaction C is

(A) $CH_3COCH_2CO_2Na$ (B) $(CH_3CO)_2O$

(C) $CH_2ClCO_2COCH_3$ (D) $CHCl_2CO_2COCH_3$

3.  Products

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Correct statement about major product(s) is :

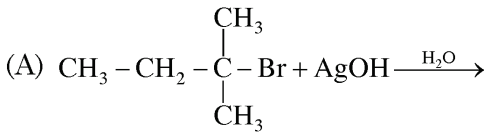
(A) It follow SN^1 and racemic mixture of alcohol is obtained

(B) It follow SN^1 and racemic mixture of alkyl iodide is obtained

(C) It follow SN^2 and racemic mixture of alcohol is obtained

(D) It follow SN^2 and racemic mixture of alkyl iodide is obtained

4. Which of the following reaction is expected to take place by S_N2 mechanism ?

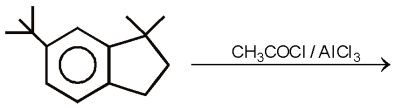
(A) 

(B) $(CH_3)_3C-Br + KOH \xrightarrow{H_2O}$

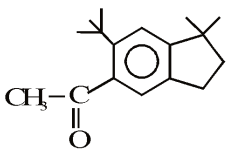
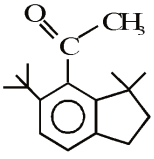
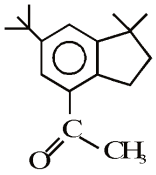
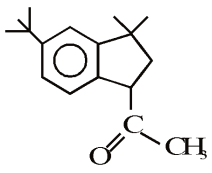
(C) $CH_3CH_2CHBrCH_3 + KOH \xrightarrow[heat]{C_2H_5OH}$

(D) $(CH_3)_2CHBr + CH_3COONa \xrightarrow{DMF}$

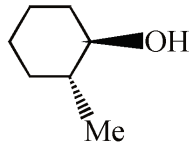
5. The major product of the reaction



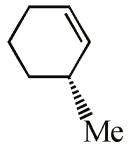
is :

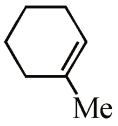
(A)  (B)  (C)  (D) 

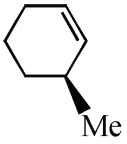
1001CJA00CHEN2500901

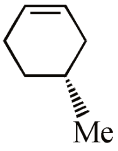
6. 

$\xrightarrow[\text{ii) Alc. KOH / }\Delta]{\text{i) } SOCl_2}$ Major product is

(A) 

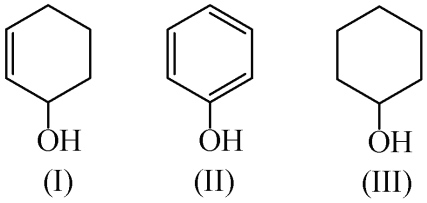
(B) 

(C) 

(D) 

7. Identify correct reactivity order for dehydration of following compounds

[Reaction with conc.H_2SO_4/Δ]-

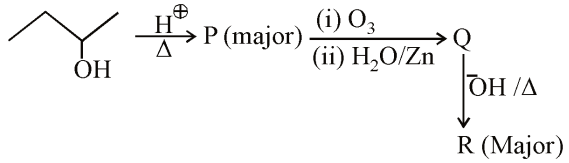


(I) (II) (III)

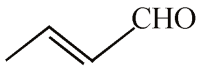
(A) I > II = III (B) I > III > II (C) III > II > I (D) II > I > III

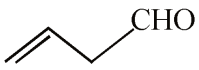
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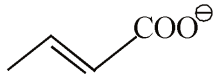
8. For the given reaction :

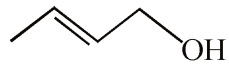


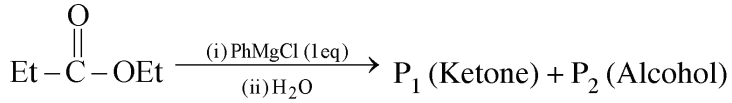
The (R) is :

(A) 

(B) 

(C) 

(D) 

9. 

True about product of above reaction

(A) P_2 will give +ve iodoform test

(B) Product P_2 will give CO_2 on heating

(C) Product P_1 will give only one oxime with NH_2OH

(D) Product P_1 will give positive Tollens test.

INTEGER TYPE

This section contains **4 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1. Number of compounds which on acidic hydrolysis gives carboxylic acid as one of the product :

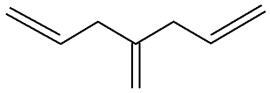
(i) CH_3CN (ii) CH_3NC (iii) CH_3CONH_2 (iv) CH_3COCl

(v) $(CH_3CO)_2O$ (vi) CH_3COOCH_3 (vii) CH_3CHO (viii) CH_3CH_2OH

(ix) CH_3COCH_3 (x) CH_3OCH_3

2. Total number of possible ether for molecular formula $C_4H_{10}O$ [Excluding Stereo Isomers]

3. 2-Methylpentane on monochlorination produces 'x' isomers. Product mixture on fractional distillation gives 'y' fractions. Then (x–y) is

4.  $\xrightarrow[\Delta]{O_3/H_2O}$ Product (A) + B (gas)

The number of moles of gas [B] per mole of given compound.

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SOT TYPE

This section contains **14 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(–1)**

1. $CH_3-CH_2-CO_2H \xrightarrow[+Br_2]{\text{red P}} \underset{\text{(optically pure form)}}{P_1} \xrightarrow{NH_3} P_2$

Which option is incorrect ?

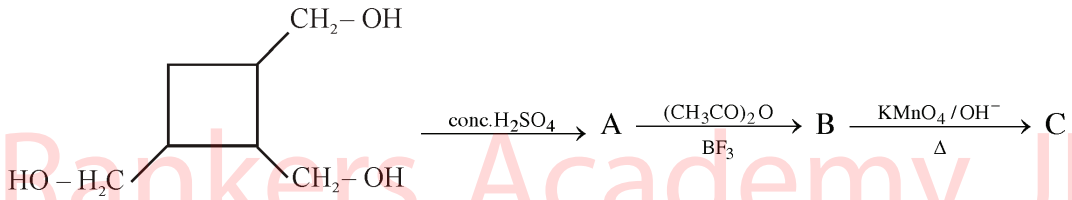
(A) P_1 gives +ve test with $NaHCO_3$

(B) P_2 optically active dipolar ion at pH = 6–7

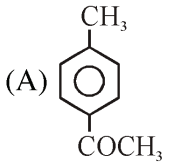
(C) Reaction involved in P_1 formation is Hell Volhard zalinsky reaction

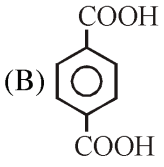
(D) P_1 is optically inactive bromo acid

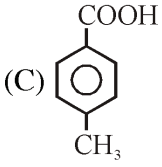
2. The final major product of the following reaction is

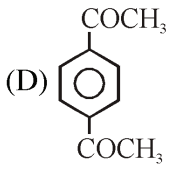


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(A) 

(B) 

(C) 

(D) 

3. The Lucas test is used to determine the type of –

(A) Amines (B) Alcohols (C) Acids (D) Phenols

4.  $+ CH_2Cl_2 + \underset{\text{(excess)}}{AlCl_3} \longrightarrow X$

$X + NBS \longrightarrow Y$

$Y \xrightarrow{\text{Hydrolysis}} Z$

Correct option regarding X, Y, Z.

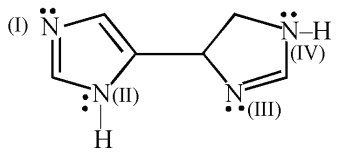
(A) Z when reacts with lucas reagent, white turbidity is formed within 5-10 minutes.

(B) Maximum number of benzenoid structures X can have is two.

(C) Z when reacts with PCC, benzophenone is formed as product.

(D) Y when reacts with silver nitrate, white precipitate is formed instant.

5. Which of the following statement is **INCORRECT** about given compound ?



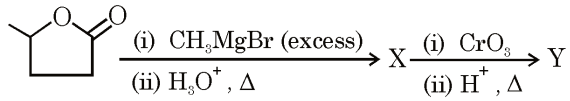
(A) Compound can not give electrophilic aromatic substitution

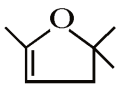
(B) Nitrogen labelled as (IV) is more basic than (II) but lesser basic than that of (I)

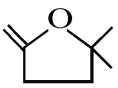
(C) Given compound is having aromatic character.

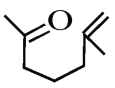
(D) Basic strength of Nitrogen (III) > Nitrogen (IV)

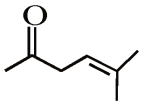
6. The major product of following reaction is



(A) 

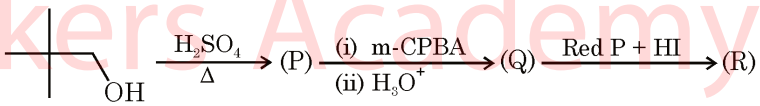
(B) 

(C) 

(D) 

7. For the given reaction

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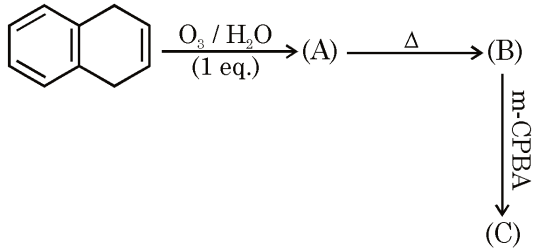
Incorrect statement is :

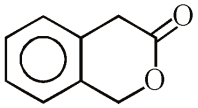
(A) Number of monohalogenated product for (R) is 6

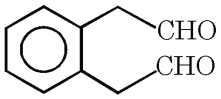
(B) (P) show positive bromine water test

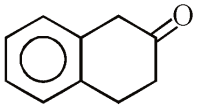
(C) (Q) gives methane with CH_3MgBr

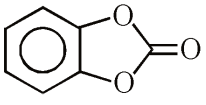
(D) (Q) does not show the reaction with Na

8. 

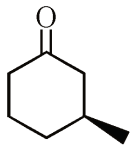
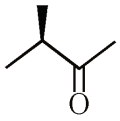
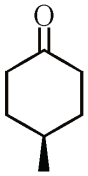
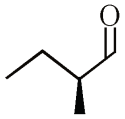
(A) Final product (C) is 

(B) Compound (A) is 

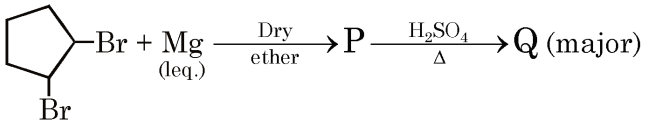
(C) Compound (B) is 

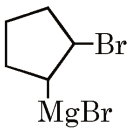
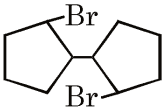
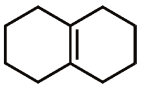
(D) Compound (C) is 

9. Which of the following will not give pair of diastereomers on reaction with CH_3MgBr followed by H_2O :

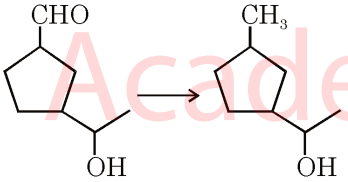
(A) 
(B) 
(C) 
(D) 

10. In the following reaction, compound (Q) is



(A) 
(B) 
(C) 
(D) 

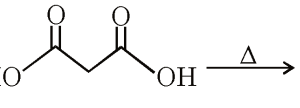
11. Best reagent to carry out following conversion is :

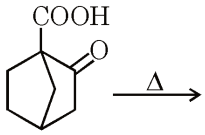


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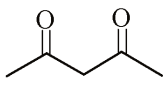
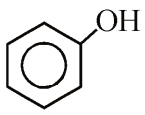
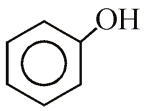
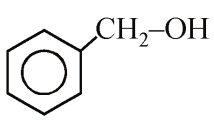
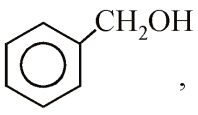
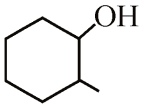
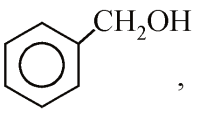
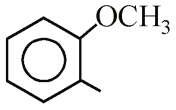
(A) Red P + HI (B) $LiAlH_4$, Et_2O (C) Zn-Hg / HCl (D) (i) N_2H_4 / $\overset{\ominus}{O}H$, Δ

12. Which of the following reaction will not produce CO_2 ?

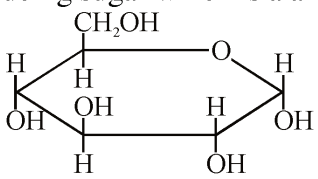
(A) $CH_3–CH_2–SO_3H \xrightarrow{NaHCO_3}$ (B) 

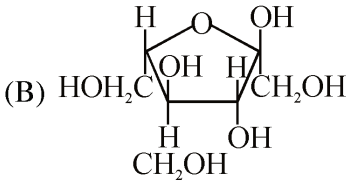
(C) $CH_3–CH_2–COONa \xrightarrow{\text{Electrolysis}}$ (D) 

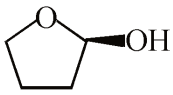
13. Which of the following can be distinguished by neutral $FeCl_3$.

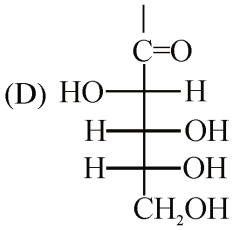
(A)  , 
(B)  , 
(C)  , 
(D)  , 

14. Reducing sugar which is a aldohexsose is :

(A) 

(B) 

(C) 

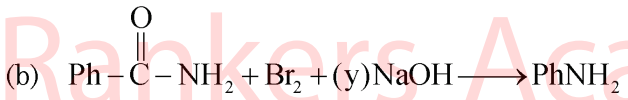
(D) 

INTEGER TYPE

This section contains **2 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1. Express your answer as x + y + z :

(a) All possible products obtained by cross cannizzaro's reaction between HCHO and PhCHO is (X).

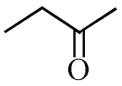
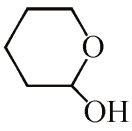
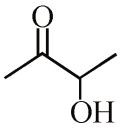
(b) 

(c) Number of resonating structure possible for phenoxide ion is (Z).

Fill your answer as sum of digits till you get the single digit answer.

2. How many of the following shows positive tollen's test

(a) HCOOH (b) HCHO (c) CH_3COOH (d) CH_3CHO

(e)  (f) Ph – CHO (g)  (h) 

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SOT TYPE : 4(–1)

1.	(B)	2.	(C)	3.	(B)	4.	(C)
5.	(A)	6.	(B)	7.	(C)	8.	(B)
9.	(B)	10.	(A)	11.	(C)	12.	(B)

INTEGER TYPE : 4(–1)

1.	(2)	2.	(2)	3.	(3)

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SOT TYPE : 4(–1)

1. (A)	2. (B)	3. (D)	4. (A)
5. (D)	6. (B)	7. (C)	8. (B)
9. (D)	10. (A)	11. (B)	12. (B)
13. (B)			

INTEGER TYPE : 4(–1)

1. (5)	2. (4 or 5)	3. (5)

Rankers Academy JEE

SOT TYPE : 4(–1)

1. (A)	2. (B)	3. (D)	4. (C)
5. (C)	6. (C)	7. (D)	8. (C)
9. (B)	10. (C)	11. (A)	12. (C)
13. (B)			

INTEGER TYPE : 4(–1)

1. (5)	2. (3)

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SOT TYPE : 4(–1)

1.	(C)	2.	(C)	3.	(C)	4.	(B)
5.	(D)	6.	(A)	7.	(C)	8.	(C)
9.	(D)	10.	(B)	11.	(D)	12.	(A)
13.	(D)	14.	(D)	15.	(D)		

INTEGER TYPE : 4(–1)

1.	(6)	2.	(4)

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SOT TYPE : 4(–1)

1.	(A)	2.	(C)	3.	(B)	4.	(A)
5.	(B)	6.	(D)	7.	(A)	8.	(D)
9.	(A)	10.	(A)	11.	(B)	12.	(C)

INTEGER TYPE : 4(–1)

1.	(5)	2.	( )	3.	(5)

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SOT TYPE : 4(–1)

1.	Ans. (A)	2.	Ans.(D)	3.	Ans.(A)	4.	Ans. (B)
5.	Ans. (D)	6.	Ans. (B)	7.	Ans. (B)	8.	Ans. (B)
9.	Ans.(D)	10.	Ans.(D)	11.	Ans. (D)	12.	Ans (D)

INTEGER TYPE : 4(–1)

1.	Ans. (4)	2.	Ans. (5)

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SOT TYPE : 4(–1)

1.	Ans. (B)	2.	Ans.(B)	3.	Ans.(A)	4.	Ans. (C)
5.	Ans. (D)	6.	Ans. (D)	7.	Ans. (A)	8.	Ans. (D)
9.	Ans.(B)	10.	Ans.(C)				

INTEGER TYPE : 4(–1)

1.	Ans. (7)	2.	Ans.(5)	3.	Ans. (4)	4.	Ans.(6)
5.	Ans.(4)						

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SOT TYPE : 4(–1)

1.	Ans. (B)	2.	Ans. (B)	3.	Ans. (C)	4.	Ans. (B)
5.	Ans. (A)	6.	Ans. (C)	7.	Ans. (D)	8.	Ans. (D)
9.	Ans. (D)	10.	Ans. (B)				

INTEGER TYPE : 4(–1)

1.	Ans. (5)	2.	Ans. (6)	3.	Ans. (4)

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SOT TYPE : 4(–1)

1.	Ans. (A)	2.	Ans. (B)	3.	Ans. (B)	4.	Ans.(D)
5.	Ans.(C)	6.	Ans. (A)	7.	Ans. (B)	8.	Ans. (A)
9.	Ans. (A)						

INTEGER TYPE : 4(–1)

1.	Ans. (6)	2.	Ans. (3)	3.	Ans.(3)	4.	Ans.(5)

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SOT TYPE : 4(–1)

1.	Ans. (D)	2.	Ans. (B)	3.	Ans.(B)	4.	Ans.(C)
5.	Ans.(A)	6.	Ans.(D)	7.	Ans. (D)	8.	Ans. (A)
9.	Ans. (B)	10.	Ans.(D)	11.	Ans. (D)	12.	Ans. (D)
13.	Ans. (B)	14.	Ans. (A)				

INTEGER TYPE : 4(–1)

1. Ans. 245 OMR ans (2)
2. Ans. (6)

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